

产品开发更改

Date / 日期: 2026.03.25

Step 1: Change request & Basic information

Component No. / 零部件号:

F03Z20046V-01

Initiator/发起人:	FENG Ying
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Reason of changes / 更改理由:

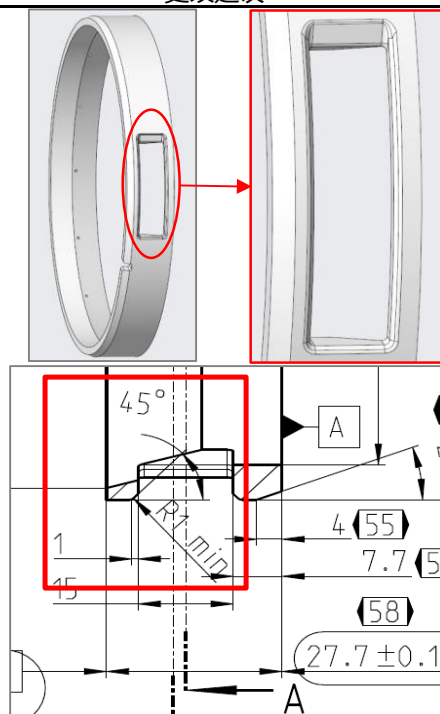
导油环回油口处未设计倒角，入机壳时导油环发生卡滞，压装不到位

Design optimization / 设计优化

Other / 其它	
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Step 2: Change proposal & Change description

备注



回油口增加C角和圆角，避免压装发生卡滞

回油口增加: C角1x45°和圆角R1

 BOSCH	Impact analysis		Edition 2.0		
Confidential		影响分析			
		Corresponding DC No. / 对应的开发更改编号: RBCE-PDECR2026006			
		Date / 日期: 2026.03.25			
Step 3.1 Impact analysis / 影响分析					
1. Function&Performance will be influenced? / 产品功能性能影响?		☒ no/否	☐ yes/是	Confirmed by: __FENG Ying__	
2. Interface and Apperance will be influenced? / 接口和外观影响?		☒ no/否	☐ yes/是	Confirmed by: __FENG Ying__	
3. Reliability and robustness will be influenced? / 产品可靠性、鲁棒性影响?		☐ no/否	☒ yes/是		
Description of above points, and concern point analysis / 对上述影响的描述, 以及分析是否存在顾虑点					
产品可靠性提高					
4. Others component will be influenced? / 影响其它零部件?		☒ no/否	☐ yes/是	Confirmed by: __FENG Ying__	
If yes, Which components will change in parallel? 如果影响, 哪些零部件也需要同时变更?					
5. Manufactory/assembly/testing will be influenced? / 加工, 装配, 测试是否影响?		☒ no/否	☐ yes/是	Confirmed by: __HU Bin__	
6. Influence on supplier part? Feedback from Supplier/Purchasing?		☒ yes/是	☐ no/否	Confirmed by: __ZHANG Lijuan__	
采购件变更					
7. Influence on System/Hardware/Software/Calibration/Mechanical parts?		☐ yes/是	☒ no/否	Confirmed by: __FENG Ying__	
e.g.: If changes influnce software/calibration, and ME/HW version alignment is required. e.g. SW reflashing affected products in stock e.g. Mechanical change or HW change triggered SW change, inform SW team e.g. System element change trigger calibration change, inform Calibration team					
8. Influence on cost / 对成本的影响		☐ increase/增加	☐ decrease/降低	☒ no change/不变	
成本不变					
9. Stock/Delivery treatment / 库存(发货)处理指示:					
Mixed Deliveries Permissible? 改前改后是否可以混合供货? ☒ YES ☐ NO					
How to deal with 1st delviery after change? 改后第一批货物的发货要求? Answer回答: __维持一样__					
Raw materials: ☒ Not affect ☐ Use in other products ☐ Scrap ☐ Rework ☐ Use up (包含在途)					
Parts/Subassemblie: ☒ Not affect ☐ Use in other products ☐ Scrap ☐ Rework ☐ Use up					
Finished goods(inhouse): ☒ Not affect ☐ Scrap ☐ Rework ☐ Use up					
Finished goods(RDC外库): ☒ Not affect ☐ Scrap ☐ Rework ☐ Use up					
Finished goods(customer): ☒ Not affect ☐ Recall ☐ Rework (包含在途)					
Step 3.2 Quality Assurance Items / 检查项目					
Validations 验证		Plan finish date 计划完成日期	Resp.person 负责人	Comments 备注	
☒ Trial Run		2026.04.02	Hu Bin	报告体现压装深度数据	
☐ Capability Studies_CMK					
☐ Capability Studies_MSA					
☐ MAE release					
☐ Cleanness test					
☐ QZ test					
☐ 200h PDL					
☐ BOM check					
☐ Test report					
☐ PAV release					
Step 3.3 Affected documents Check/ 影响文档检查					
1 Interface FMEA-relevant / IFMEA:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
2.Product FMEA-relevant / DFMEA:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
3. Special Charecteristics relevant/PSC:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
4. IMDS relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
5 Offer drawing relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
6. TCD relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
7. Norm, WB, HF... relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
8 ...		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
Step 4 Technical feasibility & Validation plan approval / 技术可行性&验证计划批准					
Development/ 研发	Purchasing/采购	MFE/工艺	Quality/质量	CPjM/客户项目	COS/样品
XIANG Liangshan	TAO Jiong	HE Yonggang	XIA Qian	SU Jian	HE Yonggang
MOEx/ 生产	LOG/物流	Others (in needed)			
LUO Zhi	XU Baochun				
* Note: if reflash SW in joint location, e.g. in ME plant, the max. 2 resp. are needed to approve the PD-ECR, e.g. MOE DM +EPQ					

Confidential		BOSCH		Implementation		Document approval: RBGD/EGC		Edition 2.0	
Confidential				导入确认					
				Corresponding DC No. / 对应的开发更改编号: RBCE-PDECR2026006				Date /日期: 2026.03.25	
Step 5 Documents release / 文档发布									
1. Documents release (drawing, offer drawing, Bom, Spec.,) / 文档发布 (图纸, 客户图纸, ...)									
Development confirmation/ 研发确认: FENG Ying									
Step 6.1 Implementation check list / 导入清单									
Departments	Y/N	Description				Responsible	Due date		
Development	N	Change BOMs & Drawings & Documents in POE system				FENG Ying			
	N	Inform documents update (check work-on can met requirements)				FENG Ying			
	Y	Update Offer drawing, TCD, D-FMEA							
	N	Norm, WB, HF...							
	N	MoC, IMDS.							
Manufacturing	Y	Related (Production/Testing) equipement be ready on site							
	Y	Related (Production/Testing) program be ready (If related to lettering, labeling or marking change, it's mandatory to provide the change evidence in Remark column)							
	Y	Related (Production/Testing) tooling / cutting / fixture etc. be ready							
	Y	Old tooling / cutting / fixture disposal							
	Y	Old materials disposal							
	Y	Planner update the planning sheet							
	Y	Update FMEA							
	Y	Update CP/FC (Control Plan/Flow Chart)							
	Y	Update WI/PDS (Include attachments.)							
	Y	First batch Mark, Special Mark (Inside Package)							
	Y	First batch Mark, Special Mark (Outside Package)							
	Y	Training							
COS	Y	Confirm the storage of old parts and coordinate the introduction date for new part (RM)							
	Y	Confirm the delivery date of old parts and first delivery of new parts (FG)							
	Y	Check sample orders which affected:material order of CKD							
	Y	Confirm production scheduling according to the alignment, any changes share the information to MOEx,MFE							
	Y	Confirm the old stock/ do prioritize delivery and inventory handling							
	Y	Inform the first delivery to PMO							
Purchasing	Y	Check sample orders which affected: material order of purchasing parts							
	Y	Inform internal related departments(COS,MFE,MOEx)with following requirements.:Yellow sticker, delivery quantities and date, PN No. and name,change description...							
	Y	Update incoming inspection plan							
Quality	Y	Update testing programe on testing equipment							
	Y	Update inspection plan for CKD parts							
CPjM	Y	Distribute the Offer drawing, TCD to customer							
LOP	Y	Check 10 digit material order							
PMO	Y	Check sample orders which affected: Customer order							
	Y	Inform Customer the first delivery information							
Others (e.g. actions in HW, SW)									
Step 6.2 Implementation date / 执行日期									
Planned implementation date: 变更计划执行日期									
Step 7 Implementation approval / 导入清单									
Development/ 研发	Purchasing/采购	MFE/工艺	COS/样品	Quality/质量	CPjM/客户项目	MOEx/生产	LOG/物流		
XIANG Liangshan	TAO Jiong	HE Yonggang	HE Yonggang	XIA Qian	SU Jian	LUO Zhi	XU Baochun		
Other/其他									
\									

Example of affected actions

Related domains
ME
Specific analysis points

HW
Specific analysis points

SW
Specific analysis points

SW
Specific analysis points

SW
Specific analysis points

PD-ECR check points

Step 3.1 Impact analysis / 影响分析

5. Manufactory/assembly/testing will be influenced? / 加工, 装配, 测试是否影响?

- whether production fixture needs to consider Poka Yoke to prevent manual assembly with wrong direction in Types change?
- whether change over check list needs to prepare for operators when type changes?
- whether the PLC program and working parameter for Screw tightening needs updating?
- whether additional assembly/testing risks by the change?

Step 3.2 Quality Assurance Items / 检查项目

- BOM check
- Test report
- PAV release

Step 3.1 Impact analysis / 影响分析

5. Manufactory/assembly/testing will be influenced? / 加工, 装配, 测试是否影响?

SW reflash operation needs to do:

- Scenario 1 (Highest efficient way): Rework at Customer side, reflash SW at EOL station or by OTA.
- Scenario 2: Return to Bosch Plant: Rework at internal area for SW reflash, key points to be defined: Rework site, Operator, ReflashTool, necessary Documents, RASIC

Step 6.1 Implementation check list / 导入清单

SW introduction to Production:

- RBCD(Calibration engineer) offline deliver updated SW code by mail to ME plant for try run
- ME upload to manufacturing SW flash station (interface?)
- ME try run with the new upgrated SW and feedback result to RBCD
- officially send upgrated SW to ME via workon during PD-ECR approval (2days)
(update SW if need based on the try run result)

Step 6.1 Implementation check list / 导入清单

e.g. Machanic electrical valvel design changed, triggered SW version updated, Prevention defined to prevent from Label/type error:

add SW identification code into Barcode for ensuring traceability



Team discussion results to be documented

Nr.	Change content	version	Date
1	Initial version	v1	Apr. 2025
2	1) Change scope cover SYS,ME, HW 2) Add linkage of SW activity	v2	Dec. 2025

Editor

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